PURELY INSEPARABLE EXTENSIONS ARE TRANSITIVE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The transitivity half of the purely inseparable machinery, companion to the purely inseparable subextension lemma (b7 card L48372).

Editorial note. The label lemma-purely-inseparable-permanence names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Chapter definition: an element alpha is purely inseparable over k when alpha raised to some p-power lies in k, p the characteristic (p = 0 case means separable, handled by the chapter convention; here p is positive)
- A product of two p-powers is a p-power

Lemma 0.1. *Let E/k and F/E be purely inseparable extensions of fields. Then F/k is a purely inseparable extension of fields.*

Proof. Say the characteristic of k is p. Choose $\alpha \in F$. Then $\alpha^q \in E$ for some p-power q. Whereupon $(\alpha^q)^{q'} \in k$ for some p-power q'. Hence $\alpha^{qq'} \in k$. □

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